Glacier Surface Area Change Detection: Quelccaya Ice Cap, Peru (1990-2024)

Summary

This project quantified long-term changes in the estimated glacier/snow-covered surface area of the Quelccaya Ice Cap in Peru using Landsat satellite imagery and Google Earth Engine. A Python-based workflow was developed to process Landsat Collection 2 Level-2 imagery, classify glacier/snow-covered pixels using the Normalized Difference Snow Index (NDSI), apply terrain-based filtering, calculate surface area, and generate visual outputs.

The results show a clear long-term decline in estimated glacier/snow-covered area from 58.53 km² in 1990 to 37.06 km² in 2024, equivalent to a total loss of approximately 21.5 km² or 36.7% (*Image 1*). The estimated rate of loss over the study period is approximately 0.63 km² per year. Although short-term fluctuations are present, the overall trend is consistent with published observations of rapid glacier retreat at Quelccaya and across the tropical Andes.

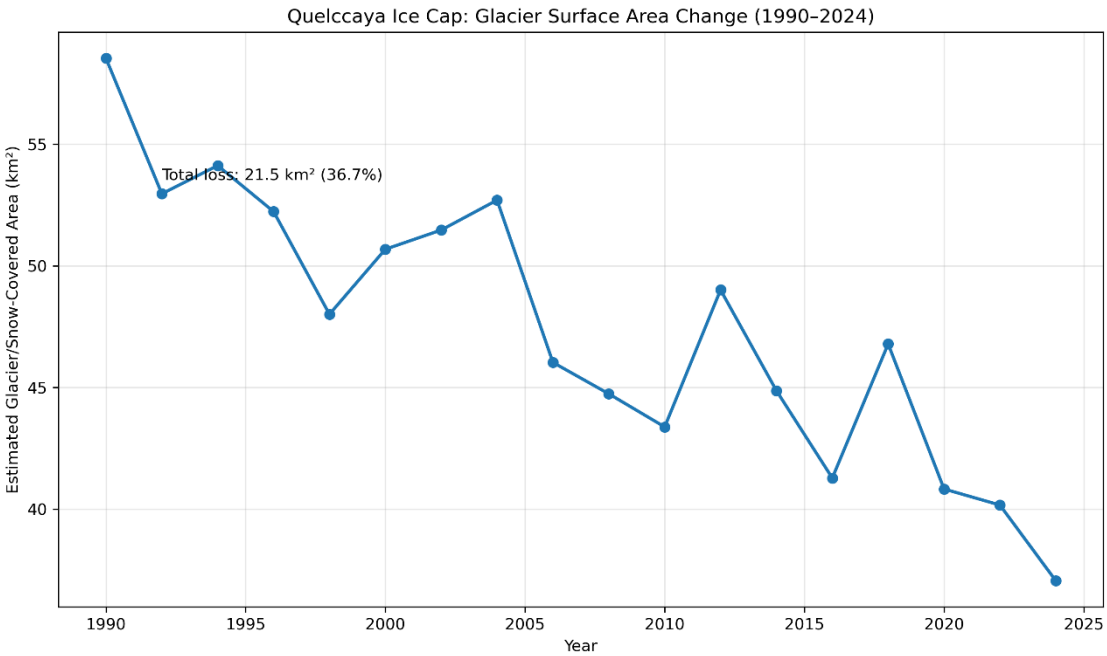

Image 1: Glacier/snow-covered area time series for the Quelccaya Ice Cap, 1990-2024

1. Introduction

Tropical glaciers are highly sensitive indicators of climate change because they exist close to melting conditions and respond rapidly to small shifts in temperature, precipitation, and freezing-level height. The Andes contain nearly all of the world's tropical ice cover, making Andean glaciers particularly important for monitoring tropical climate change and its impacts on water resources (Albert et al., 2014). The

Quelccaya Ice Cap, located in the Cordillera Vilcanota of southern Peru, is one of the most important and widely studied tropical ice bodies.

The scientific importance of Quelccaya is well established. Previous research describes it as the largest single ice cap in the tropics and highlights its sensitivity to changes in the elevation of the 0°C isotherm due to its dome-shaped geometry and high-altitude setting. Studies also note that the glacier margins include a range of topographic gradients, slopes, and aspects, making the ice cap as a whole a useful target for studying glacier–climate interactions rather than focusing only on a single outlet glacier.

Recent literature shows that tropical Andean glaciers have been retreating rapidly for several decades. This retreat has important implications for dry-season water availability, natural hazards, and climate records preserved in ice. Vuille et al. (2018) describe tropical glaciers as sentinels of rapid climate change and emphasize that glacier meltwater is especially important during the long dry season in the southern tropical Andes. NASA Earth Observatory (2023) similarly highlights the retreat of Quelccaya between 1988 and 2023 and notes that the ice cap feeds the Vilcanota River and Lake Sibinacocha, making it relevant for nearby and downstream populations.

The objective of this project was to use open satellite imagery and cloud-based geospatial processing to estimate glacier/snow-covered area change at Quelccaya between 1990 and 2024.

2. Study Area

The study focused on the Quelccaya Ice Cap in the Cordillera Vilcanota, southern Peru. The region of interest used in the analysis was defined as a polygon covering the main ice cap and its immediate surroundings:

[-70.90, -14.00]
[-70.70, -14.00]
[-70.70, -13.83]
[-70.90, -13.83]

This area captures the main Quelccaya ice body and surrounding terrain required for spatial context. The Cordillera Vilcanota is located in southeastern Peru and experiences a marked seasonal climate, with a wet/warm season from approximately October/November to March/April and a dry/cold season from April/May to September/October. Hanshaw and Bookhagen (2014) describe the

region as part of the central Andes in southeastern Peru and note that most precipitation occurs during the warm/wet season, while ablation can occur year-round due to high solar radiation.

3. Data

The analysis used Landsat Collection 2 Level-2 Surface Reflectance imagery accessed through Google Earth Engine. Different Landsat missions were used depending on the year:

Period	Sensor
1990-1998	Landsat 5
1999-2012	Landsat 7
2013-2024	Landsat 8

Images were filtered between 1 August and 30 October for each selected year. This period was chosen to reduce seasonal snow contamination and improve the likelihood of observing stable glacier margins. This decision is consistent with previous glacier-mapping approaches at Quelccaya. Albert et al. (2014) report that images used to reconstruct Quelccaya ice extent were mostly selected during the dry season, when ephemeral snow cover is lower and the ice cap margins are more visible.

A digital elevation model was also used to apply terrain-based filtering. The script used the SRTM DEM available in Google Earth Engine.

4. Methodology

4.1. Image Filtering and Preprocessing

For each year, Landsat imagery was filtered by:

- Region of interest
- Date window: 1 August to 30 October
- Cloud cover threshold: less than 20%

A median composite was generated for each year to reduce noise from individual images. Landsat Level-2 optical bands were scaled using the standard reflectance transformation:

$$Reflectance = DN \times 0.0000275 - 0.2$$

The use of multispectral satellite imagery for glacier area mapping is well established. Hanshaw and Bookhagen (2014) used a large archive of multispectral

imagery to produce glacier and lake outlines in the Cordillera Vilcanota, including the Quelccaya Ice Cap.

4.2. Glacier/Snow-Covered Area Detection

Glacier/snow-covered pixels were detected using the Normalized Difference Snow Index (NDSI):

$$NDSI = (Green - SWIR) / (Green + SWIR)$$

Band selection depended on the Landsat sensor:

Sensor	Green Band	SWIR Band
Landsat 5/7	SR_B2	SR_B5
Landsat 8	SR_B3	SR_B6

An NDSI threshold of 0.45 was applied. Pixels above this threshold were classified as glacier/snow-covered candidates. This type of visible/SWIR-based glacier mapping is consistent with earlier remote-sensing work, where visible-to-SWIR ratios were found to be among the most effective automated methods for delineating glacier extent (Albert et al., 2014).

4.3. Terrain-Based Filtering

To reduce misclassification from seasonal snow and steep terrain, three additional filters were applied:

Parameter	Value
NDSI threshold	>0.45
Elevation threshold	>5100 m
Slope threshold	< 50°
Minimum connected pixels	>40 pixels

The elevation threshold was designed to remove lower-elevation snow patches unlikely to represent persistent glacier cover. The slope threshold reduced misclassification on steep, shaded, or illuminated terrain. Connected-pixel filtering removed small isolated clusters that were unlikely to represent meaningful glacier-covered areas.

This approach is a simplified but transparent threshold-based classification. It does not attempt to separate permanent ice from seasonal snow perfectly. For this reason, results are reported as estimated glacier/snow-covered area, not as a definitive glacier inventory.

4.4. Area Calculation

The binary glacier/snow-covered mask was multiplied by pixel area and aggregated over the study region using Google Earth Engine's "reduceRegion" function. The result was converted from square meters to square kilometers.

The workflow produced a CSV table containing:

- Year
- Estimated glacier/snow-covered area in km²
- Percent change from the previous observation

4.5. Visual Outputs

Three visual outputs were generated:

1. A time-series plot of estimated glacier/snow-covered area.
2. A true-color before/after map comparing 1990 and 2024, with the classified glacier/snow-covered area overlaid in cyan.
3. A GIF animation showing the evolution of the classified glacier/snow-covered mask over time.

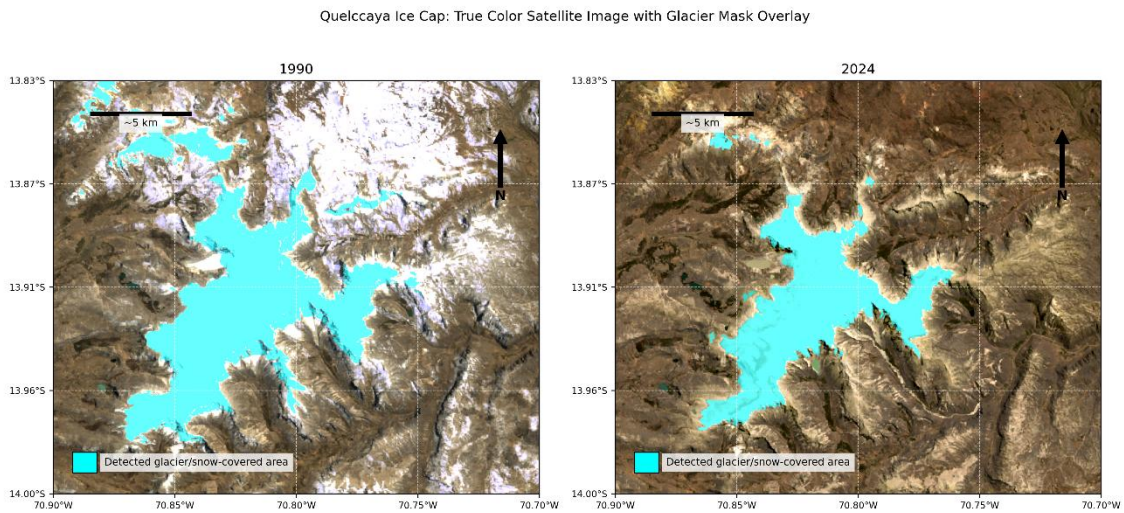

Image 2: True-color Landsat comparison of Quelccaya Ice Cap in 1990 and 2024

5. Results

The estimated glacier/snow-covered area declined from 58.53 km² in 1990 to 37.06 km² in 2024. This represents a total estimated loss of 21.47 km², or approximately 36.7% of the 1990 area.

The long-term trend is clearly negative, although the time series shows interannual variability. Some observations show temporary increases in estimated

glacier/snow-covered area. These short-term increases should not be interpreted as true glacier growth. They are more likely caused by residual seasonal snow, cloud effects, sensor differences, and threshold sensitivity.

The overall result is broadly consistent with independent observations. NASA Earth Observatory (2023) reports that Quelccaya's ice area was approximately 58 km^2 in 1988 and just over 40 km^2 in 2023, based on Landsat imagery. The present analysis estimates 58.53 km^2 in 1990 and 37.06 km^2 in 2024, which is close in magnitude and direction, although the 2024 value is slightly lower due to the specific classification parameters and region of interest used in this project. The estimated average loss rate in this project is approximately 0.63 km^2/year. This is also comparable to Hanshaw and Bookhagen's (2014) reported decline rate of 0.57 ± 0.10 km^2/year for the Quelccaya Ice Cap over the 1980–2010 period.

6. Spatial Interpretation

The before/after comparison shows that the detected glacier/snow-covered area was substantially larger and more continuous in 1990 than in 2024. By 2024, the classified area appears more restricted around the central ice cap, with reduced extent around the margins. The strongest visible retreat occurs along peripheral areas, consistent with the expectation that glacier margins are more sensitive to warming, seasonal snow loss, and changes in the freezing level.

The spatial pattern is also consistent with previous mapping work. Albert et al. (2014) reported that retreat occurred along all margins of the Quelccaya Ice Cap when comparing satellite-derived extents from 1990 and 2009. NASA's 1988–2023 image comparison similarly shows a substantial reduction in the ice edge and the appearance or growth of meltwater lakes along the ice cap margin.

7. Comparison with Literature

The results align with several published estimates of Quelccaya and tropical Andean glacier retreat.

Albert et al. (2014) report that the Quelccaya Ice Cap retreated from 58.9 km^2 in 1975 to 40.8 km^2 in 2010, a loss of 31%, or approximately 0.5 km^2/year. This project estimates a loss of 36.7% from 1990 to 2024. The higher percentage loss is reasonable because the study period extends further into the recent period, during which continued retreat has been widely documented.

Hanshaw and Bookhagen (2014) found that the Quelccaya Ice Cap declined at 0.57 ± 0.10 km^2/year between 1980 and 2010, while also noting that lower-elevation glaciers in the Cordillera Vilcanota retreat faster than higher-elevation glaciers. The estimated loss rate in this project, 0.63 km^2/year, falls close to this published value.

More recent work by Taylor et al. (2022) shows that glacier retreat remains widespread in the southern Peruvian Andes and reports continued upward shifts in glacier median elevation, including a 15.9 m per decade increase over the Quelccaya Ice Cap between 1975 and 2020. This supports the interpretation that the observed area reduction in this project reflects a broader pattern of retreat and elevation-driven glacier change.

The wider regional context also supports the results. Vuille et al. (2018) describe a situation of rapidly shrinking glacier area, length, and volume across the tropical Andes, particularly since the late 1970s. They also highlight the role of increasing temperature and interannual climate variability, including ENSO, in shaping glacier mass balance.

8. Limitations

This analysis provides a useful and reproducible estimate of long-term glacier/snow-covered area change, but several limitations must be acknowledged.

First, NDSI detects snow and ice together. The classified area should therefore be interpreted as estimated glacier/snow-covered area rather than pure glacier ice. Seasonal snow can persist into the dry season, especially after snowfall events, leading to overestimation in some years.

Second, the method is sensitive to thresholds. The selected values for NDSI, elevation, slope, and connected-pixel filtering influence the absolute area estimates. However, testing different parameter combinations showed that the long-term declining trend remains consistent.

Third, the use of different Landsat sensors introduces additional uncertainty. Previous work on Quelccaya has also noted that variability in ice extent estimates may result from different satellite sensors, spatial resolutions, ephemeral snow, illumination conditions, and debris-covered ice margins (Albert et al., 2014).

Fourth, the analysis does not use manually digitized glacier outlines or a formal glacier inventory such as Global Land Ice Measurements from Space (GLIMS) or

Randolph Glacier Inventory (RGI). A polygon-based glacier inventory would provide a more rigorous boundary for validation.

Finally, this analysis estimates surface area only. It does not estimate volume or mass balance. Area loss is important, but glacier volume change would provide a more complete assessment of hydrological and climatic impact.

9. Conclusion

This project demonstrates how Landsat imagery, Google Earth Engine, and Python can be combined to quantify glacier/snow-covered area change over multiple decades. The Quelccaya Ice Cap showed a strong long-term decline from 58.53 km^2 in 1990 to 37.06 km^2 in 2024, equivalent to an estimated loss of 21.47 km^2 or 36.7%.

The magnitude and direction of this result are consistent with NASA observations and peer-reviewed research on Quelccaya and the tropical Andes. While short-term variability exists due to seasonal snow, threshold sensitivity, and sensor differences, the overall retreat signal is robust.

The study highlights the value of cloud-based geospatial analysis for environmental monitoring and provides a reproducible workflow that could be extended to other glaciers or used as a foundation for more advanced methods, such as machine-learning classification or glacier inventory validation.

10. Future Work

Future improvements could include:

- Comparing results against official glacier outlines from GLIMS or RGI.
- Testing multiple NDSI and elevation thresholds as a formal sensitivity analysis.
- Adding temporal smoothing to distinguish long-term trend from short-term classification noise.
- Estimating glacier volume or elevation change using DEM differencing.
- Incorporating climate variables such as temperature, precipitation, and ENSO indices.
- Replacing threshold-based classification with a supervised machine-learning classifier.

11. Bibliography

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